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Economic evaluations of musculoskeletal physiotherapy: Protocol of a systematic review

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Economic evaluations of musculoskeletal physiotherapy: Protocol of a systematic review

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Abstract:

Introduction:

Several economic evaluations of musculoskeletal physiotherapy have been published in the literature. We aim to provide an overview of these existing economic evaluations. This overview will be useful for health care funders in identifying studies matching their context. In addition, research gaps as well as literature extensive enough to be combined in a meta analyses will be identified. This will support reserachers in planning relevant research projects.

Methods and Analyses:

We will search in PubMed, EconLit, and NHS-EED for relevant literature. Full economic evaluations of musculoskeletal physiotherapy interventions will be included, regardless of type, and economic evaluation perspective. Initial searches were performed on the 7th of October 2021. Study selection, data extraction, and the quality evaluation will be conducted initially by two independent researchers. If their agreement is sufficient, one reviewer will proceed with the respected process. From the included studies, we will extract information on the publication year, the country of origin, the type of economical evaluation analyses, and the specific musculoskeletal condition. An overview will be provided, concerning the distributions of the included studies regarding the extracted information. Furthermore, an evaluation of the individual study quality will be offered.

Ethics and dissemination:

No ethical approval will be required for this systematic review, since no human participants are involved. We aim to distribute the findings of this review in a peer reviewed journal, on national and international conferences, as well as via social media.

Strength and Limitations of the study:

- Inclusion of full economic evaluations only, secures relevance of included studies for health care funders
- First overview of existing economic evaluations of musculoskeletal physiotherapy interventions
- Evaluation of the included studies quality
- This systematic review will highlight research gabs in economic evaluations of musculoskeletal physiotherapy interventions

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Background:

Rationale:

Health economic evaluations extend evidence-based medicine, by considering the monetary costs besides the potential risks and benefits of an intervention (1). The limitation of financial resources for health care – including rehabilitation services – make these evaluations essential for decision makers, who try to allocate resources effectively. The scarcity in monetary resources is likely to further increase due to the medical technologies process and the aging population, which underlines the importance of economic evaluations and summarizing reviews.

Full economic evaluations are recommended as basis for health care decisions. They compare the costs and effects of at least two treatment options (1). Full economic evaluations can be categorised into cost-effectiveness, cost-utility, and cost-benefit analyses. Some economists, consider the cost utility analyses as part of cost the effectiveness analyses. The outcome of a cost effectiveness is a specific clinical outcome such as survival time. In cost utility analyses the outcome are usually quality adjusted life years (QALY), which may be considered as a specific clinical outcome. In cost benefit analyses, health effects of an intervention are monetary quantified. Thus, the willingness-to-pay is considered.

To determine the costs for full economic evaluations two different perspectives can be chosen. The health system perspective considers only direct costs involving e.g., hospital costs and out-patient services. The society perspective respects indirect costs in addition to the direct costs involving e.g., loss in work efficiency.

Physiotherapy contributes to the financial health care burden. The duty of physiotherapists is to assess, plan, and implement rehabilitation programs (2). The most important contributor to the global rehabilitation need are musculoskeletal health conditions including fractures, low back pain, and osteoarthritis (3). Thus, physiotherapy contributes to the financial burden arising from the treatment of musculoskeletal conditions and needs to be economically evaluated.

Some full economic evaluations of musculoskeletal physiotherapy as well as related systematic reviews have been performed (4, 5). Nevertheless, *for several musculoskeletal conditions* there is still a *lack of full economic evaluations of physiotherapeutic interventions*. Besides, transferability from existing evaluations depends largely on a matching design and context between the study setting and the destination. The studies' quality is further of importance to decide whether the results are transferable. Finally, the quality also informs about the extent of research gaps.

A recent systematic review of economic evaluations of neurological physiotherapeutic interventions has addressed this challenge (6). In contrast to standard systematic reviews that article only provides an overview of existing studies, based on their geographical location and underlying health condition in the neurological physiotherapeutic field. Unfortunately, the quality of the included studies was not evaluated, which would have provided valuable insights on the quality of the evidence and the research gaps. Nonetheless, the results of that

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review on the one hand help health care funders to identify relevant articles matching their destination, and on the other hand it highlights knowledge gaps and areas with sufficient articles for systematic review studies.

A similar overview of economic evaluations of musculoskeletal physiotherapy interventions is however missing. The first step to provide such an overview of existing studies and their quality is conducting a systematic review. This includes the publication of a study protocol, which secures high transparency in the research process (7).

Objective:

The aim of this systematic review is to identify studies of economic evaluations of musculoskeletal physiotherapeutic interventions.

Thus, our population of interest compromises all patients with a musculoskeletal condition, which are eligible for a physiotherapeutic treatment. The outcomes need to be derived from a full economic evaluation study. Finally, a control group needs to be involved in the included studies.

Methods:

To design this protocol and the related systematic review we follow the five-steps approach for preparing a systematic review of economic evaluations for Clinical Practice Guidelines (7). Furthermore, we utilized the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2015 statement to report this protocol (8).

Eligibility criteria:

We will include full economic evaluation studies of musculoskeletal physiotherapy, published in any language up to 2021.

Details on the inclusion and exclusion criteria are provided in Table 1. They are in line with the PICOS acronym of our research question (P= population, I= intervention, C= control, O= outcomes, S= study type).

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Table 1. Criteria for inclusion and exclusion of articles

	Inclusion criteria	**Exclusion criteria**
Population	Patients, with a primary musculoskeletal condition	Patients with intellectual disabilities and studies with heterogenous participant groups, of which the majority has different primary conditions than musculoskeletal
Intervention	Any physiotherapeutic intervention including amongst other patient education, single or group-based exercise therapy, and manual therapy	Interventions offered from non-health professionals including sport instructors and E-interventions
Control	Any control group is considered, including those treated with a wait and see approach, with a placebo or an alternative treatment.	
Outcome	cost-effectiveness ratios, cost utility ratios and net benefits	
Study type	Full economic evaluations based on clinical studies or based on modelling	Studies without results eg. Protocols Review studies

Information Sources:

We will search for relevant articles in the following databases from their inception:

- Medline (trough PubMed)
- EconLit
- NHS-EED (can only be search up to March 2015)

Initial searches in all databases were performed on the 7th October 2021. The databases EconLit and NHS-EED were chosen to secure coincide economic study inclusion, although NHS-EED is no longer updated and therefore only includes articles up to March 2015. The references list of all included studies as well as of relevant systematic literature reviews identified through the search will be screened for additional eligible studies. Finally, experts in the field will be asked for further eligible studies.

Search strategy:

After the initial search in PubMed, LB will draft a search strategy, which will be revaluated, discussed, and adjusted in the research team by utilizing the Peer Review of Electronic Search Strategies 2015 checklist (9).

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The search strategy combines three main terms “economic evaluation”, “physiotherapy”, and “musculoskeletal” (Table 2). For each of the main term’s synonyms and/or relevant examples were identified and added with an OR to the respected main term in the search. To reduce search terms and maintaining sensitivity of the search truncation was used eg. physiotherap*. “Musculoskeletal” is considered to define the included population, “Physiotherapy” describes the intervention, and “Economic evaluation” designate the outcome of interest.

Table 2. Overview of included search terms

Population	**Intervention**	**Outcome**
musculoskeletal	Physiotherap*	Cost effectiveness
chirur*	Exercise training	Cost utility
orthop*	Exercise therapy	Cost benefit*
osteoarthritis	Conservative treatment	Economic analy*
Back pain	Conservative management	Cost analy*
	Conservative therap*	Economic evaluation
	physical rehabilitation	
	Orthopaedic rehabilitation	

No limits are planned for the publication date. In the first step there will also be no limits regarding the language, however studies published in another language than English, German or Danish will be excluded during the full text screening. The search will be conducted in “all fields” of the articles, including title and abstract as well as the main body text.

To secure, that the review will be up to date, prior to submission, the search will be redone and potential, new released, relevant articles will be incorporated.

Data management:

The articles identified via the searches in the different databases will be merged in EndNote X9. Articles included for full-text assessment will be stored in Endnote as well. To extract study characteristics and findings, an excel spread sheet will be prepared in accordance with Broman and Woo (10).

Selection process:

At first, duplicates will be removed automatically via Endnote. Remaining duplicates will be sorted out by hand. Title and abstracts of identified articles will be screened by two independent reviewers (LB und BK). In case of disagreement a discussion will be initiated if no consensus can be reached a third reviewer (AH or HHK) will be involved. Full texts of all articles, deemed to be relevant from first title-abstract screening, will be inspected by

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the same two reviewers. They will decide on articles final inclusion in the systematic review. Disagreement will be solved via discussion and consulting of a third reviewer, if necessary (AH or HHK). The reason for each study excluded, based on full text reading will be recorded and incorporated in the systematic review. The references of all included studies as well as of identified relevant systematic reviews will be checked for additional eligible studies. In- and exclusion of articles will be displayed in a flow-chart.

Data collection process:

Prior to the data extraction LB will extract the data of two random studies to evaluate the user-friendliness and completion of the data extraction sheets, and potential adjustments will be executed. Afterwards, data from three studies will be pilot extracted by two reviewers (LB and BK). Discrepancies will be discussed until consensus will be reached and if necessary, the spreadsheet will be revised. If the agreement of the data extraction is high one reviewer will extract the data from the remaining studies. In case of insufficient agreement, the pilot extraction will be repeated. Authors of studies with missing data will be contacted via e-mail to complete the spreadsheet. A reminder will be sent out two weeks after the initial contact. Unresolved uncertain or missing data will be marked as such in the review.

Data items:

Information on the data items extracted from the included studies may be found in table 3.

Table 3. Overview of the data items to be extracted

Item	**Content of the item**
Study ID	First author´s last name and year of publication
Registration	If available
Location	Country or countries where the investigation was conducted
Setting/Context	Inpatient vs. outpatient and prevention/prehabilitation vs rehabilitation
Study design	Randomized control trials vs. modelling
Population	Number of included participants, including their mean age and sex distribution
Pathology/Condition	Eg. fracture vs. osteoarthritis vs. rheumatology vs. back pain
Type of intervention	Single vs. group based Exercises therapy vs. education vs. manual therapy vs. combination of therapies Length of intervention

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	Unites of interventions
Control intervention	Wait-and-see vs. usual care vs. surgical procedure
Effect measure	Eg. QALY vs. years of life vs. walking speed vs. pain vs. physical functioning
Time horizon	Length of follow up to calculate effects and costs in month
Type economic analysis	cost-benefit vs. cost-utility vs. cost-effectiveness
Cost Perspective	Health system perspective vs. societal perspective
Discounting	Yes vs. no/not stated
Inflation rate	Yes vs. no/not stated
Reference year and currency	Year of data collection, if not stated it will be highlighted but also approximated Eg. US Dollar vs. Euros
Missing data	No missing data vs. not stated vs. percentage of missing data and information of imputation or not
Analyses to handle statistical and other uncertainties	Statistical uncertainties: Confidence intervals for Incremental cost-effectiveness ratios and cost-effectiveness acceptability curve Other uncertainties: Sensitivity analyses Yes vs. No, and if yes which
Source of funding	Profit vs. non-profit vs. mixed vs. unclear vs. no funding In case of funding extraction of funders name
Competing interests	Yes vs. No vs. not stated If yes: extract specification
Authors conclusion	A summary of the authors conclusion In favour of intervention vs. in favour of control vs. uncertain

EHR: Electronical Health Records; QALY: quality-adjusted life-years

Outcomes and prioritisation:

The main outcome of interest is the descriptive overview of existing economic evaluations of musculoskeletal physiotherapy interventions.

Risk of bias in and methodological quality of individual studies:

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The quality of the individual economic evaluations will be investigated with the Consensus on Health Economic Criteria checklist (11). Two reviewers (LB and BK) will independently evaluate three of the included studies. Discrepancies will be discussed until consensus is reached. If their initial agreement is high, one reviewer will evaluate the remaining studies. If their agreement is insufficient, they will evaluate another 3 studies and repeat the procedure.

Data Synthesis:

The results of the included studies will be synthesized by different characteristics. We will provide an overview of the publication years, countries of origin, type of analyses, and the specific musculoskeletal condition.

Metabias(es):

We will investigate if available associated protocols deviate from the results of the included studies, to address reporting bias. Furthermore, we will search for study protocols and trial registrations at the ISRCTN registry matching our study aim, to account for a publication bias. An overview of these protocols and trial registrations will be added to the systematic review.

Confidence in cumulative evidence and transferability:

Since we do not combine the results to formulate conclusions or suggestions, it is not applicable for this type of review to provide a confidence measure on the transferability.

Study dates:

The systematic review is currently ongoing. Preliminary searches and conception started on the 1st September 2021. The anticipated date for review completion is the 1. June 2022.

Patient and Public involvement:

Patient public involvement was not involved in designing, conducting, reporting, or disseminating this study.

Ethics and dissemination:

Ethics approval is not required since no human participants or their identifiable information was contained in this study. To follow good research practice for systematic reviews we submitted a PROSPERO protocol on the 7th October 2021. The findings of the reviewers are anticipated to be presented at international conferences, published in a peer reviewed journal, and disseminated via social media.

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Amendments:

Amendments from this protocol will be documented in the related PROSPERO protocol.

Discussion:

The aim of our upcoming systematic review is to provide an overview of existing economic evaluation studies of musculoskeletal physiotherapy and to evaluate their quality. This paper presents the protocol of this systematic review.

The protocol aids preventing waste of research resources. It informs the research community about this ongoing work and may prevent, that additional resources are spent on this topic only duplicating the results. Furthermore, publishing a systematic review protocol supports research transparency and is in line with the recommendations from van Mastrigt *et al.* on the preparation of a systematic review of economic evaluations (7).

This systematic review will provide an overview of existing full economic evaluations of musculoskeletal physiotherapy interventions. This overview will be helpful to health care funders searching for economic evaluations of musculoskeletal physiotherapy matching their specific context of interest.

Moreover, researchers will benefit from this review results as knowledge gaps as well as areas with sufficient literature for a meta-analysis will be highlighted, which may inspire and guide future research. Our review might, for example, identify that most economic evaluations are performed in Europe, whereas there might be a lack of studies from the African countries. Another example could be that several cost-effectiveness studies on physiotherapeutic interventions for knee osteoarthritis may already exist but that there is still a lack of cost utility analyses. A final example of a result of our review could be that, despite the existence of cost effectiveness-studies for the physiotherapeutic treatment after a lower limb fracture, there would be still a need for further investigations due to the poor quality in sampling of the existing studies.

A strength of the review is the inclusion of only full economic evaluations since it secures relevance of the included studies for health care funders.

Nonetheless, systematic reviews have their limitations, especially raising from the quality of the methods used. It needs to be distinguished between limitations within the systematic review and within the included studies.

Publication bias is one of the within review concerns. Studies which have been performed but were not published eg. due to negative results, may lead to a waste of additional research resources. To address this issue, we will add information to our review on relevant ongoing studies and published protocols, which we find during our search process.

The quality of included studies in meta-analyses is of importance since poor quality of the original studies also weakens the reliability of the review. Therefore, we aim not only for providing an overview of the existing

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economic evaluations of musculoskeletal physiotherapy but also to evaluate their quality. This can assist in identifying areas with limited quality of the evidence.

Contributors:

Linda Baumbach: Conceptualization, Methodology, Writing- Original Draft, Writing – Review & Editing.

Benedikt Kretzler: Conceptualization, Methodology, Writing- Original Draft, Writing – Review & Editing.

Hans-Helmut König: Conceptualization, Methodology, Writing- Original Draft, Writing – Review & Editing.

André Hajek: Conceptualization, Methodology, Writing – Review & Editing.

All authors read and approved the final manuscript.

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Competing interests: None.

Patient consent for publication: Not required.

Supplementary material: None

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PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item
ADMINISTRATIVE INFORMATION		
Title:		
Identification	1a	Identify the report as a protocol of a systematic review ***Economic evaluations of musculoskeletal physiotherapy: Protocol of a systematic review***
Update	1b	If the protocol is for an update of a previous systematic review, identify as such ***No update***
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number ***We submitted our registration to PROSPERO and currently wait for their feedback***
Authors:		
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author ***Baumbach, Linda1; König, Hans-Helmut1; Kretzler, Benedikt1; Hajek, André1*** ***1 Department of Health Economics and Health Services Research, University Medical Center Hamburg-Eppendorf, GermanyLinda Baumbach*** ***Department of Health Economics and Health Services Research,*** ***University Medical Center Hamburg-Eppendorf,*** ***Martinistr. 52, 20246 Hamburg, Germany*** ***Telephone: +49 40 7410 59047, Email: l.baumbach@uke.de***
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review ***Linda Baumbach: Conceptualization, Methodology, Writing- Original Draft, Writing – Review & Editing.*** ***Benedikt Kretzler: Conceptualization, Methodology, Writing- Original Draft, Writing – Review & Editing.*** ***Hans-Helmut König: Conceptualization, Methodology, Writing- Original Draft, Writing – Review & Editing.*** ***André Hajek: Conceptualization, Methodology, Writing – Review & Editing.***

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		All authors read and approved the final manuscript.
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments ***Amendments from this protocol will be documented in the related PROSPERO protocol.***
Support:		
Sources	5a	Indicate sources of financial or other support for the review ***none***
Sponsor	5b	Provide name for the review funder and/or sponsor ***none***
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol ***Not applicable***
INTRODUCTION		
Rationale	6	Describe the rationale for the review in the context of what is already known ***Check see page 3 background***
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO) ***The aim of this systematic review is to identify studies of economic evaluations of musculoskeletal physiotherapeutic interventions.*** ***Thus, our population of interest compromises all patients with a musculoskeletal condition, which are eligible for a physiotherapeutic treatment. The outcomes need to be derived from a full economic evaluation study. Finally, a control group needs to be involved in the included studies.***
METHODS		
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review ***We will include full economic evaluation studies of musculoskeletal physiotherapy, published in any language up to***

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		2021. ***Details on the inclusion and exclusion criteria are provided in Table 1.***
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage ***Databases: Medline (trough PubMed, EconLit, and NHS-EED*** ***For further details see page 5 Information Sources***
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated ***See page 5 and 6 Search strategy including table 2 with the included search terms***
Study records:		
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review ***Endnote X9 and Excel spread sheats***
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis) ***Two independent reviewers will select the studies for further details see page 6 Selection process***
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators ***Pilot extracting to test user-friendliness, independent extraction by two reviews to test agreement and decide n next step see Data collection process on page 7***
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications ***See Table 3, which provides an overview of all information to be extracted***
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale ***The main outcome of interest is the descriptive overview of existing economic evaluations of musculoskeletal physiotherapy interventions.***

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Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis ***The quality of the individual economic evaluations will be investigated with the Consensus on Health Economic Criteria checklist. Applied by two reviewers.***
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised ***Data will be synthesized by different characteristics to provide the overview of existing economical evaluations of musculoskeletal physiotherapy interventions.***
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ) ***Not applicable***
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression) ***Not applicable***
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned ***Not applicable***
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies) ***We will investigate if available associated protocols deviate from the results of the included studies, to address reporting bias. Furthermore, we will search for study protocols and trial registrations at the ISRCTN registry matching our study aim, to account for a publication bias. An overview of these protocols and trial registrations will be added to the systematic review.***
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE) ***This is not applicable as we do not aim to provide a overall evidence based recommendation, but an overview of existing studies***

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

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From: Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart L, PRISMA-P Group. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. BMJ. 2015 Jan 2;349(jan02 1):g7647.

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Primary Subject Heading:	Health economics
Secondary Subject Heading:	Rehabilitation medicine, Health services research
Keywords:	HEALTH ECONOMICS, SPORTS MEDICINE, THERAPEUTICS, Rehabilitation medicine < INTERNAL MEDICINE

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Economic evaluations of musculoskeletal physiotherapy: Protocol of a systematic review

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Abstract:

Introduction:

Several economic evaluations of musculoskeletal physiotherapy have been published in the literature. We aim to provide an overview of these existing economic evaluations. This overview will be useful for health care funders in identifying studies matching their context. In addition, research gaps as well as literature extensive enough to be combined in a meta analyses will be identified. This will support reserachers in planning relevant research projects.

Methods and Analyses:

We will search in PubMed, EconLit, and NHS-EED for relevant literature. Full economic evaluations of musculoskeletal physiotherapy interventions will be included, regardless of type, and economic evaluation perspective. Initial searches were performed on the 7th of October 2021. Study selection, data extraction, and the quality evaluation will be conducted initially by two independent researchers. If their agreement is sufficient, one reviewer will proceed with the respected process. From the included studies, we will extract information on the publication year, the country of origin, the type of economical evaluation analyses, and the specific musculoskeletal condition. An overview will be provided, concerning the distributions of the included studies regarding the extracted information. Furthermore, an evaluation of the individual study quality will be offered.

Ethics and dissemination:

No ethical approval will be required for this systematic review, since no human participants are involved. We aim to distribute the findings of this review in a peer reviewed journal, on national and international conferences, as well as via social media.

Strength and Limitations of the study:

- *This systematic review protocol enhances the quality of the following research work by reducing the chance of duplication and by providing transparency to the methods and procedures*
- *Study inclusion and study quality will be evaluated – both independently by two reviewers*
- *Our search is restricted to three electronic databases and restricted to German and English language*

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Background:

Rationale:

Health economic evaluations extend evidence-based medicine, by considering the monetary costs besides the potential risks and benefits of an intervention (1). The limitation of financial resources for health care – including rehabilitation services – make these evaluations essential for decision makers, who try to allocate resources effectively. The scarcity in monetary resources is likely to further increase due to the medical technologies process and the aging population, which underlines the importance of economic evaluations and summarizing reviews.

Full economic evaluations are recommended as basis for health care decisions. They compare the costs and effects of at least two treatment options (1). Full economic evaluations can be categorised into cost-effectiveness, cost-utility, and cost-benefit analyses. Some economists, consider the cost utility analyses as part of cost the effectiveness analyses. The outcome of a cost effectiveness is a specific clinical outcome such as survival time. In cost utility analyses the outcome are usually quality adjusted life years (QALY), which may be considered as a specific clinical outcome. In cost benefit analyses, health effects of an intervention are monetary quantified. Thus, the willingness-to-pay is considered.

To determine the costs for full economic evaluations two different perspectives can be chosen. The health system perspective considers only direct costs involving e.g., hospital costs and out-patient services. The society perspective respects indirect costs in addition to the direct costs involving e.g., loss in work efficiency.

Physiotherapy contributes to the financial health care burden. The duty of physiotherapists is to assess, plan, and implement rehabilitation programs (2). The most important contributor to the global rehabilitation need are musculoskeletal health conditions including fractures, low back pain, and osteoarthritis (3). Thus, physiotherapy contributes to the financial burden arising from the treatment of musculoskeletal conditions and needs to be economically evaluated.

Some full economic evaluations of musculoskeletal physiotherapy as well as related systematic reviews have been performed (4, 5). Nevertheless, *for several musculoskeletal conditions* there is still a *lack of full economic evaluations of physiotherapeutic interventions*. Besides, transferability from existing evaluations depends largely on a matching design and context between the study setting and the destination. The studies' quality is further of importance to decide whether the results are transferable. Finally, the quality also informs about the extent of research gaps.

A recent systematic review of economic evaluations of neurological physiotherapeutic interventions has addressed this challenge (6). In contrast to standard systematic reviews that article only provides an overview of existing studies, based on their geographical location and underlying health condition in the neurological physiotherapeutic field. Unfortunately, the quality of the included studies was not evaluated, which would have provided valuable insights on the quality of the evidence and the research gaps. Nonetheless, the results of that

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review on the one hand help health care funders to identify relevant articles matching their destination, and on the other hand it highlights knowledge gaps and areas with sufficient articles for systematic review studies.

A similar overview of economic evaluations of musculoskeletal physiotherapy interventions is however missing. The first step to provide such an overview of existing studies and their quality is conducting a systematic review. This includes the publication of a study protocol, which secures high transparency in the research process (7).

Objective:

The aim of this systematic review is to identify studies of economic evaluations of musculoskeletal physiotherapeutic interventions.

Thus, our population of interest compromises all patients with a musculoskeletal condition, which are eligible for a physiotherapeutic treatment. The outcomes need to be derived from a full economic evaluation study. Finally, a control group needs to be involved in the included studies.

Methods:

To design this protocol and the related systematic review we follow the five-steps approach for preparing a systematic review of economic evaluations for Clinical Practice Guidelines (7). Furthermore, we utilized the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2015 statement to report this protocol (8).

Eligibility criteria:

We will include full economic evaluation studies of musculoskeletal physiotherapy, published in any language up to the 31st of January 2022.

Details on the inclusion and exclusion criteria are provided in Table 1. They are in line with the PICOS acronym of our research question (P= population, I= intervention, C= control, O= outcomes, S= study type).

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Table 1. Criteria for inclusion and exclusion of articles

	Inclusion criteria	Exclusion criteria
Population	Patients, with a primary musculoskeletal condition	Patients with intellectual disabilities and studies with heterogenous participant groups, of which the majority has different primary conditions than musculoskeletal
Intervention	Any intervention, which can be conducted by a physiotherapist alone, including amongst other patient education, single or group-based exercise therapy, and manual therapy, any additional treatment would need to be offered/delivered in the intervention and in the control group (eg. pain medication or surgery)	Interventions offered from non-health professionals including sport instructors and E-interventions or by an interdisciplinary team, including studies in which the physiotherapeutic treatment of interest is combined with treatments mostly delivered from other disciplines
Control	Any control group is considered, including those treated with a wait and see approach, with a placebo or an alternative treatment.	
Outcome	cost-effectiveness ratios, cost utility ratios and net benefits	
Study type	Full economic evaluations based on clinical studies or based on modelling	Conference abstracts and studies without results eg. Protocols, Review studies

Information Sources:

We will search for relevant articles in the following databases from their inception:

- Medline (through PubMed)
- EconLit
- NHS-EED (can only be search up to March 2015)

Initial searches in all databases were performed on the 7th October 2021. The databases EconLit and NHS-EED were chosen to secure coincide economic study inclusion, although NHS-EED is no longer updated and therefore only includes articles up to March 2015. The references list of all included studies as well as of relevant systematic literature reviews identified through the search will be screened for additional eligible studies. Finally, experts in the field will be asked for further eligible studies.

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Search strategy:

After the initial search in PubMed, LB will draft a search strategy, which will be revaluated, discussed, and adjusted in the research team by utilizing the Peer Review of Electronic Search Strategies 2015 checklist (9).

The search strategy combines three main terms “economic evaluation”, “physiotherapy”, and “musculoskeletal” (Table 2). For each of the main term’s synonyms and/or relevant examples were identified and added with an OR to the respected main term in the search. To reduce search terms and maintaining sensitivity of the search truncation was used eg. physiotherap*. “Musculoskeletal” is considered to define the included population, “Physiotherapy” describes the intervention, and “Economic evaluation” designate the outcome of interest. A detailed description of the search is provided in supplementary file 1.

Table 2. Overview of included search terms

Population	**Intervention**	**Outcome**
musculoskeletal	Physiotherap*	Cost effectiveness
chirur*	Exercise training	Cost utility
orthop*	Exercise therapy	Cost benefit*
osteoarthritis	Conservative treatment	Economic analy*
Back pain	Conservative management	Cost analy*
	Conservative therap*	Economic evaluation
	physical rehabilitation	
	Orthopaedic rehabilitation	

No limits are planned for the publication date. In the first step there will also be no limits regarding the language, however studies published in another language than English, German or Danish will be excluded during the full text screening. The search will be conducted in “all fields” of the articles, including title and abstract as well as the main body text.

To secure, that the review will be up to date, prior to submission, the search will be redone and potential, new released, relevant articles will be incorporated.

Data management:

The articles identified via the searches in the different databases will be merged in EndNote X9. Articles included for full-text assessment will be stored in Endnote as well. To extract study characteristics and findings, an excel spread sheet will be prepared in accordance with Broman and Woo (10).

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Selection process:

At first, duplicates will be removed automatically via Endnote. Remaining duplicates will be sorted out by hand. Title and abstracts of identified articles will be screened by two independent reviewers (LB und BK). In case of disagreement a discussion will be initiated if no consensus can be reached a third reviewer (AH or HHK) will be involved. Full texts of all articles, deemed to be relevant from first title-abstract screening, will be inspected by the same two reviewers. They will decide on articles final inclusion in the systematic review. Disagreement will be solved via discussion and consulting of a third reviewer, if necessary (AH or HHK). The reason for each study excluded, based on full text reading will be recorded and incorporated in the systematic review. The references of all included studies as well as of identified relevant systematic reviews will be checked for additional eligible studies. In- and exclusion of articles will be displayed in a flow-chart.

Data collection process:

Prior to the data extraction LB will extract the data of two random studies to evaluate the user-friendliness and completion of the data extraction sheets, and potential adjustments will be executed. Afterwards, data from three studies will be pilot extracted by two reviewers (LB and BK). Discrepancies will be discussed until consensus will be reached and if necessary, the spreadsheet will be revised. If the agreement of the data extraction is high one reviewer will extract the data from the remaining studies. In case of insufficient agreement, the pilot extraction will be repeated. Authors of studies with missing data will be contacted via e-mail to complete the spreadsheet. A reminder will be sent out two weeks after the initial contact. Unresolved uncertain or missing data will be marked as such in the review.

Data items:

Information on the data items extracted from the included studies may be found in table 3.

Table 3. Overview of the data items to be extracted

Item	**Content of the item**
Study ID	First author´s last name and year of publication
Registration	If available
Location	Country or countries where the investigation was conducted
Setting/Context	Inpatient vs. outpatient and prevention/prehabilitation vs rehabilitation
Study design	Randomized control trials vs. modelling
Population	Number of included participants, including their mean age and sex distribution

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Pathology/Condition	Eg. fracture vs. osteoarthritis vs. rheumatology vs. back pain
Type of intervention	Single vs. group based Exercises therapy vs. education vs. manual therapy vs. combination of therapies Length of intervention Unites of interventions
Control intervention	Wait-and-see vs. usual care vs. surgical procedure
Effect measure	Eg. QALY vs. years of life vs. walking speed vs. pain vs. physical functioning
Time horizon	Length of follow up to calculate effects and costs in month
Type economic analysis	cost-benefit vs. cost-utility vs. cost-effectiveness
Cost Perspective	Health system perspective vs. societal perspective
Discounting	Yes vs. no/not stated
Inflation rate	Yes vs. no/not stated
Reference year and currency	Year of data collection, if not stated it will be highlighted but also approximated Eg. US Dollar vs. Euros
Missing data	No missing data vs. not stated vs. percentage of missing data and information of imputation or not
Analyses to handle statistical and other uncertainties	Statistical uncertainties: Confidence intervals for Incremental cost-effectiveness ratios and cost-effectiveness acceptability curve Other uncertainties: Sensitivity analyses Yes vs. No, and if yes which
Source of funding	Profit vs. non-profit vs. mixed vs. unclear vs. no funding In case of funding extraction of funders name
Competing interests	Yes vs. No vs. not stated If yes: extract specification
Authors conclusion	A summary of the authors conclusion In favour of intervention vs. in favour of control vs. uncertain

EHR: Electronical Health Records; QALY: quality-adjusted life-years

Outcomes and prioritisation:

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The main outcome of interest is the descriptive overview of existing economic evaluations of musculoskeletal physiotherapy interventions.

Risk of bias in and methodological quality of individual studies:

The quality of the individual economic evaluations will be investigated with the Consensus on Health Economic Criteria checklist (11). Two reviewers (LB and BK) will independently evaluate three of the included studies. Discrepancies will be discussed until consensus is reached. If their initial agreement is high, one reviewer will evaluate the remaining studies. If their agreement is insufficient, they will evaluate another 3 studies and repeat the procedure.

Data Synthesis:

The results of the included studies will be synthesized by different characteristics. We will provide an overview of the publication years, countries of origin, type of analyses (clinical trial vs. modelling with a stratification for decision tree, Markov model, discrete event simulation), time of interventions (primary, secondary, or tertiary prevention), over all types of interventions (primary or add-on, eg. after surgery), and the specific musculoskeletal condition. A special focus will be set on the latter one, i.e., the summary of results of the included studies including a stratification for specific type of intervention will be provided by musculoskeletal condition.

Metabias(es):

We will investigate if available associated protocols deviate from the results of the included studies, to address reporting bias. Furthermore, we will search for study protocols and trial registrations at the ISRCTN registry matching our study aim, to account for a publication bias. An overview of these protocols and trial registrations will be added to the systematic review.

Confidence in cumulative evidence and transferability:

Since we do not combine the results to formulate conclusions or suggestions, it is not applicable for this type of review to provide a confidence measure on the transferability.

Study dates:

The systematic review is currently ongoing. Preliminary searches and conception started on the 1st September 2021. The anticipated date for review completion is the 1. June 2022.

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Patient and Public involvement:

Patient public involvement was not involved in designing, conducting, reporting, or disseminating this study.

Ethics and dissemination:

Ethics approval is not required since no human participants or their identifiable information was contained in this study. To follow good research practice for systematic reviews we submitted a PROSPERO protocol on the 7th October 2021 (CRD42021276050). The findings of the reviewers are anticipated to be presented at international conferences, published in a peer reviewed journal, and disseminated via social media.

Amendments:

Amendments from this protocol will be documented in the related PROSPERO protocol (CRD42021276050).

Discussion:

The aim of our upcoming systematic review is to provide an overview of existing economic evaluation studies of musculoskeletal physiotherapy and to evaluate their quality. This paper presents the protocol of this systematic review.

The protocol aids preventing waste of research resources. It informs the research community about this ongoing work and may prevent, that additional resources are spent on this topic only duplicating the results. Furthermore, publishing a systematic review protocol supports research transparency and is in line with the recommendations from van Mastrigt *et al.* on the preparation of a systematic review of economic evaluations (7).

This systematic review will provide an overview of existing full economic evaluations of musculoskeletal physiotherapy interventions. This overview will be helpful to health care funders searching for economic evaluations of musculoskeletal physiotherapy matching their specific context of interest.

Moreover, researchers will benefit from this review results as knowledge gaps as well as areas with sufficient literature for a meta-analysis will be highlighted, which may inspire and guide future research. Our review might, for example, identify that most economic evaluations are performed in Europe, whereas there might be a lack of studies from the African countries. Another example could be that several cost-effectiveness studies on physiotherapeutic interventions for knee osteoarthritis may already exist but that there is still a lack of cost utility analyses. A final example of a result of our review could be that, despite the existence of cost effectiveness-studies for the physiotherapeutic treatment after a lower limb fracture, there would be still a need for further investigations due to the poor quality in sampling of the existing studies.

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A strength of the review is the inclusion of only full economic evaluations since it secures relevance of the included studies for health care funders.

Nonetheless, systematic reviews have their limitations, especially raising from the quality of the methods used. It needs to be distinguished between limitations within the systematic review and within the included studies.

Publication bias is one of the within review concerns. Studies which have been performed but were not published eg. due to negative results, may lead to a waste of additional research resources. To address this issue, we will add information to our review on relevant ongoing studies and published protocols, which we find during our search process.

The quality of included studies in meta-analyses is of importance since poor quality of the original studies also weakens the reliability of the review. Therefore, we aim not only for providing an overview of the existing economic evaluations of musculoskeletal physiotherapy but also to evaluate their quality. This can assist in identifying areas with limited quality of the evidence.

Contributors:

Linda Baumbach: Conceptualization, Methodology, Writing- Original Draft, Writing – Review & Editing.

Benedikt Kretzler: Conceptualization, Methodology, Writing- Original Draft, Writing – Review & Editing.

Hans-Helmut König: Conceptualization, Methodology, Writing- Original Draft, Writing – Review & Editing.

André Hajek: Conceptualization, Methodology, Writing – Review & Editing.

All authors read and approved the final manuscript.

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Competing interests: None.

Patient consent for publication: Not required.

Supplementary material: None

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Search strategies

Searches were performed in the following databases:

1. Medline (trough PubMed)
2. EconLit
3. NHS-EED (can only be search up to March 2015)

There were no filters or limits applied the search.

The following search was performed in each of the three databases:

((((economic analy*) OR (cost analy*) OR (cost benefit*) OR (cost utility) OR (cost effectiveness) OR (economic evaluation))) AND (((muskuloskeletal) OR (chirur*) OR (orthop*) OR (osteoarthritis) OR (back pain)))) AND (((orthopaedic rehabilitation) OR (physical rehabilitation) OR (exercise therap*) OR (conservative therap*) OR (conservative management) OR (conservative treatment) OR (exercise training) OR (physiotherap*)))

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Reporting checklist for protocol of a systematic review.

Based on the PRISMA-P guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

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In your methods section, say that you used the PRISMA-Preporting guidelines, and cite them as:

Moher D, Shamseer L, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart LA. Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2015 statement. Syst Rev. 2015;4(1):1.

		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1 (line 1,2)
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	Not applicable
Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	10 (line 205,206, 210)
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	11(line245-249)
Amendments			

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	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	Not applicable
Support			
Sources	#5a	Indicate sources of financial or other support for the review	11(line 250)
Sponsor	#5b	Provide name for the review funder and / or sponsor	Not applicable
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	Not applicable
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is already known	3
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	4
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	5
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	5
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	6
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	6-7
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	7

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Study records - data collection process	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	7-8
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	7-8
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	8-9
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	9
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	9
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	9
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	Not applicable
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	Not applicable
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	9
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	Not applicable

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